

Ordinary Least Squares

Lecture 2 (Stata code)

Table of contents

Reading	2
Data	3
LCF (2023)	3
Estimation: An intuitive approach	3
Guess 1	4
Guess 2	6
Structure	6
Fit	7
Aggregation	8
Minization	8
Ordinary Least Squares	8
Problem	8
First order conditions	9
Simple case	9
Sample Analogue	11
Decomposition	11
Fitted values	11
Interpretation	12
Example: LCF (2023)	12
Estimation	12
Decomposition	12
Fitted values	13
Residual	14
Binary regressor:	14
Summary stats	14

OLS - dummy variable	15
Sample analogue	16
Multivariate models	16
OLS - multivariate	16
Interpretation	17
Partialling out	17
Demonstration	18
Goodness-of-Fit	19
Measure of fit	20
Chasing R^2	20
Adjusted R^2	21
Example: LCF (2023)	21
Adding “irrelevant” variables	21
Does OLS work?	22
Accuracy	23
Random variable	23
Example: LCF (2023) - Version 2	24
Example: LCF (2023) - Version 3	24
Example: LCF (2023) - Version 4	24
References	25
Bibliography	25

Reading

This lecture covers material from Wooldridge (2025):

- Chapter 2.2 - “Deriving the Ordinary Least Squares Estimates”
- Chapter 2.3 - “Properties of OLS on Any Sample of Data”
- Chapter 2.4 - “Units of Measurement and Functional Form” (covered in Lecture 1)
- Chapter 3.2 - “Mechanics and Interpretation of Ordinary Least Squares”
- Chapter 6.3 - “More on Goodness-of-Fit and Selection of Regressors”

The lecture does not cover Chapter 2.6 “Regression through the Origin and Regression on a Constant”. I would suggest reading the section, but we will not have time to cover it.

Data

For each observation we observe:

$$[y_i, x_{i1}, x_{i2}, \dots, x_{ik}] \quad i = 1, \dots, n$$

Stacked together, we have a matrix of data:

$$\begin{bmatrix} y_1 & x_{11} & x_{12} & \dots & x_{1k} \\ y_2 & x_{21} & x_{22} & \dots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ y_n & x_{n1} & x_{n2} & \dots & x_{nk} \end{bmatrix}$$

Just like in Stata/R

LCF (2023)

Suppose you wanted to estimate the the marginal propensity to consume in the UK using data from the [Living Costs and Food Survey](#):

- a *household level* dataset with information on consumption, income, and other characteristics.

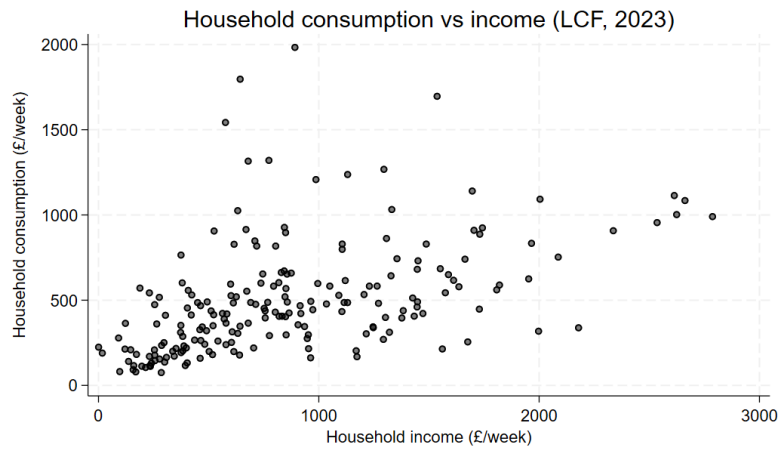
$$[cons_i, inc_i, children_i, size_i, tv_license_i, month_i] \quad i = 1, \dots, n$$

Estimation: An intuitive approach

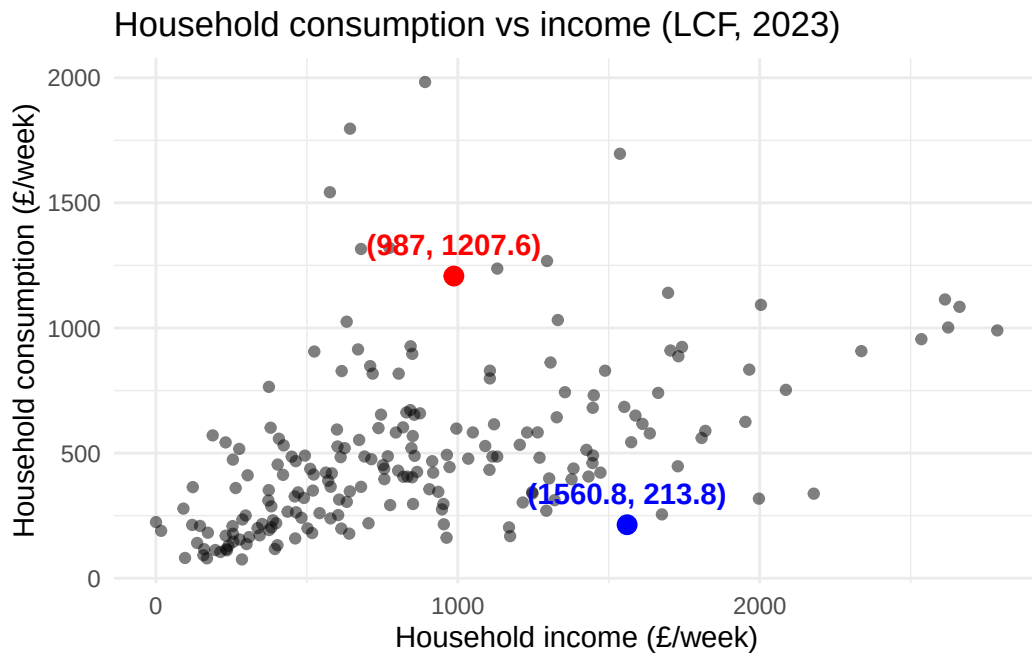
Does the relationship between consumption and income look linear?

```
* Open data
use "$data_dir\lcf_2023.dta", clear

* Plot
scatter hhcon hhinc, mc(black) mfc(%50) ytitle("Household consumption (£/week)") xtitle("Hou
```

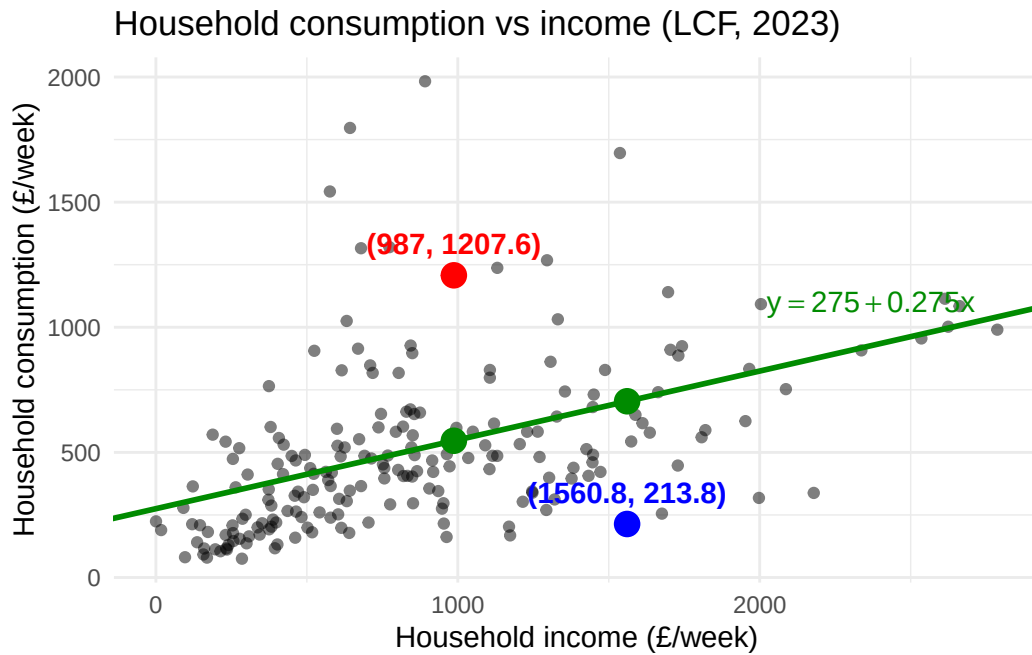


Consider the following two observations:

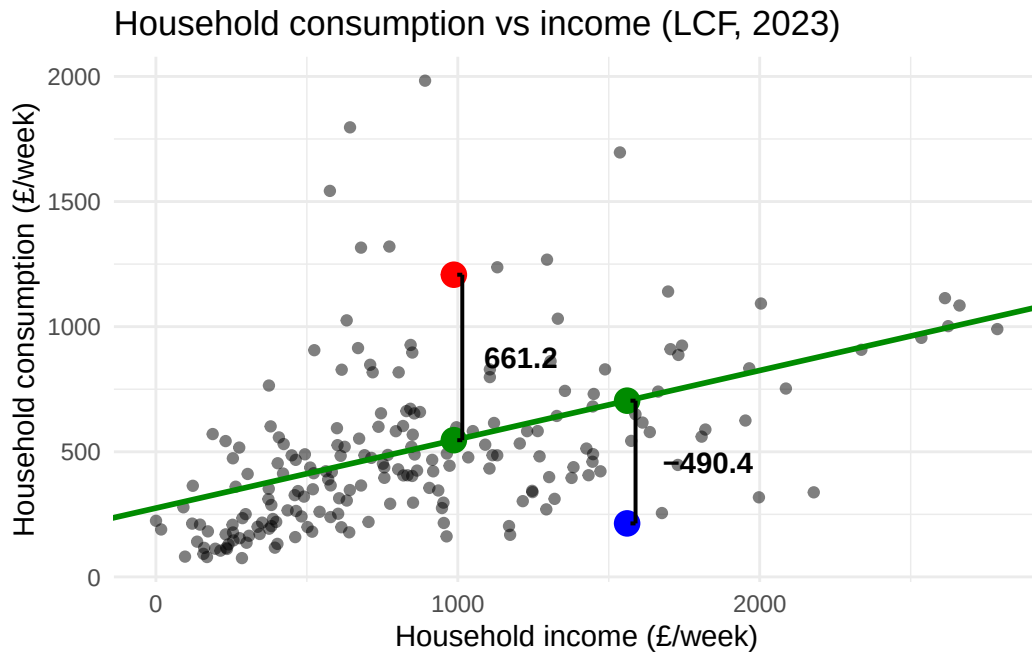


Guess 1

Suppose we plot a line through the points:

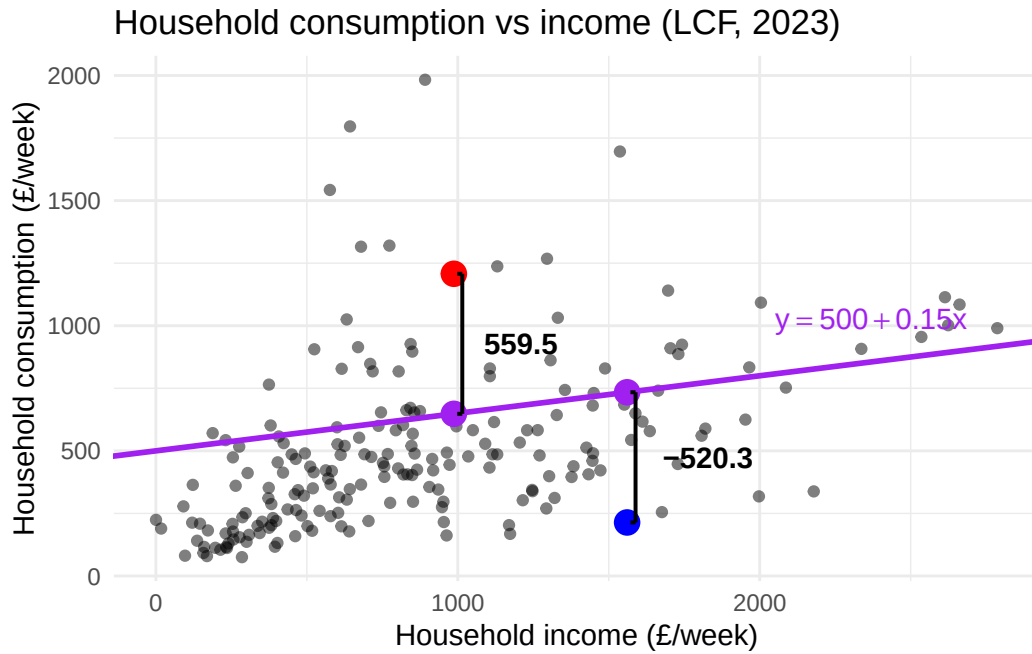


The vertical distance between points in the data and the line tells us how the difference between the “prediction” of the line (for a given inc_i) and the actual data.



Guess 2

What if we chose a different plot?



Which guess was better? Why?

Structure

Let's formalize the above process:

- **Goal:** we want to pick the line that best fits the data.
- **Problem:** what do we mean by “fit”?
 - For a given value of inc_i in the data, the “fitted” value of $cons_i$ is

$$\widehat{cons}_i = b_0 + b_1 inc_i$$

where $[b_0, b_1]$ are the chosen parameters.

– For example,

$$* \widehat{cons}_i = 275 + 0.275inc_i$$

$$* \widehat{cons}_i = 500 + 0.15inc_i$$

Fit

We need a measure of distance:

Candidate 1: difference

$$cons_i - \widehat{cons}_i$$

- **PROBLEM:** could be negative (i.e. not a measure of distance)

Candidate 2: absolute value

$$|cons_i - \widehat{cons}_i|$$

- **PRO:** intuitive
- **CON:** difficult to work with (i.e. not globally differentiable)

Candidate 3: squared deviation

$$(cons_i - \widehat{cons}_i)^2$$

- always positive
- amplifies bigger deviations, which is more *efficient* (more on that soon!)

Aggregation

What we care about is the difference between the prediction and data across all observations:

$$\sum_{i=1}^n (cons_i - \widehat{cons}_i)^2$$

And since $\widehat{cons}_i = b_0 + b_1 inc_i$

$$\sum_{i=1}^n (cons_i - b_0 - b_1 inc_i)^2$$

Minization

The value of these squared deviations (from the prediction) depend on $[b_0, b_1]$. It seems intuitive then to pick the values that minimize this value:

$$\min_{b_0, b_1} \sum_{i=1}^n (cons_i - b_0 - b_1 inc_i)^2$$

Ordinary Least Squares

Problem

Minimization problem:

$$\hat{\beta} = \arg \min_b \sum_{i=1}^n (y_i - b_0 - b_1 x_{i1} - \dots - b_k x_{ik})^2$$

where $\hat{\beta} = [\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k]$ and $b = [b_0, b_1, \dots, b_k]$.

- hence the name “Least Squares”

First order conditions

Differentiate w.r.t. β_j 's and then set to 0. For $\hat{\beta}_0$ (the constant):

$$\begin{aligned} 0 &= 2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) \\ \Rightarrow 0 &= \frac{1}{n} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) \\ \Rightarrow \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x}_1 - \dots - \hat{\beta}_k \bar{x}_k \end{aligned}$$

Note

The inclusion of a constant in the model ensures that the residual is always mean zero:

$$0 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) = \sum_{i=1}^n \hat{u}_i$$

For $\hat{\beta}_j \quad j = 1, \dots, k$

$$\begin{aligned} 0 &= 2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) x_{ij} \\ \Rightarrow 0 &= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) x_{ij} \end{aligned}$$

There are k of these first-order conditions (plus the 1 for the constant)

$\Rightarrow k + 1$ equations and $k + 1$ unknowns.

Simple case

In the case of a single regressor, the solution is more easily solved.

- For $\hat{\beta}_0$ (the constant):

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

- For $\hat{\beta}_1$:

$$\begin{aligned}
0 &= 2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) x_i \\
\Rightarrow 0 &= \sum_{i=1}^n (y_i - (\bar{y} - \hat{\beta}_1 \bar{x}) - \hat{\beta}_1 x_i) x_i \quad \text{substitute in } \hat{\beta}_0 \\
\Rightarrow 0 &= \sum_{i=1}^n ((y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x})) x_i
\end{aligned}$$

The next step exploits the fact that:

$$\sum_{i=1}^n \hat{u}_i = 0 \Rightarrow \bar{x} \sum_{i=1}^n \hat{u}_i = 0 \Rightarrow \sum_{i=1}^n \hat{u}_i \bar{x} = 0$$

Therefore we can minus 0 from the right-hand side without changing anything.

$$\begin{aligned}
0 &= \sum_{i=1}^n ((y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x})) x_i - \sum_{i=1}^n \hat{u}_i \bar{x} \\
\Rightarrow 0 &= \sum_{i=1}^n ((y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x})) (x_i - \bar{x}) \\
\Rightarrow \hat{\beta}_1 &= \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\widehat{Cov}(y_i, x_i)}{\widehat{Var}(x_i)}
\end{aligned}$$

💡 Linear algebra

The above expression - along with its multivariate counterpart - has a very nice linear algebra counterpart. The vector of OLS estimators (including the intercept) is given by,

$$\hat{\beta} = (X'X)^{-1}X'Y$$

Where X is a $n \times (k + 1)$ matrix of regressors (including vector of 1's for the intercept) and Y a $n \times 1$ vector. It's a very simple expression and demonstrates the value of knowing a little linear algebra. This how you define the OLS estimator in graduate-level modules like [EC5207](#) *Econometric Methods and Applications*.

Sample Analogue

Recall from Lecture 1, under the MLR assumptions (notably, $E[x_i u_i] = 0$)

$$\beta_1 = \frac{Cov(y_i, x_i)}{Var(x_i)}$$
$$\beta_0 = E[y_i] - \beta_1 E[x_i]$$

The OLS estimators for a simple model are the **sample analogue**

$$\hat{\beta}_1 = \frac{\widehat{Cov}(y_i, x_i)}{\widehat{Var}(x_i)}$$
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

OLS replaces expectations with sample averages.

Decomposition

Given the OLS estimates, we can decompose the outcome into two parts: fitted values ($\hat{y}_i$) and residual ($\hat{u}_i$)

$$y_i = \underbrace{\hat{\beta}_0 + \hat{\beta}_1 x_i}_{\hat{y}_i} + \hat{u}_i$$

Fitted values

You can show,

- $\sum_{i=1}^n \hat{u}_i = 0$: residuals are mean zero
- $\sum_{i=1}^n x_i \hat{u}_i = 0$: residuals and regressors are uncorrelated
- $\sum_{i=1}^n \hat{y}_i \hat{u}_i = 0$: residuals and fitted values are uncorrelated
- $\sum_{i=1}^n \hat{y}_i = \sum_{i=1}^n y_i$: fitted values have the same mean as outcome
- $\sum_{i=1}^n y_i \hat{u}_i = \sum_{i=1}^n (\hat{y}_i + \hat{u}_i) \hat{u}_i = \sum_{i=1}^n \hat{u}_i^2$

Some of these are exercises in [Tutorial 1](#)

Interpretation

The $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ also gives us the interpretation of the estimator:

$$\hat{\beta}_1 = \frac{d\hat{y}_i}{dx_i} \quad \text{OR} \quad \Delta\hat{y}_i = \hat{\beta}_1 \Delta x_i$$

and

$$\hat{\beta}_0 = \hat{y}_i \quad \text{for } x_i = 0$$

Example: LCF (2023)

Estimation

```
* Estimate bivariate linear regression
regress hhcon hhinc
```

Source	SS	df	MS	Number of obs	=	200
Model	4842359.39	1	4842359.39	F(1, 198)	=	57.08
Residual	16798378.5	198	84840.2954	Prob > F	=	0.0000
Total	21640737.9	199	108747.427	R-squared	=	0.2238
				Adj R-squared	=	0.2198
				Root MSE	=	291.27

hhcon	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
hhinc	.2705895	.0358165	7.55	0.000	.1999587	.3412203
_cons	268.0611	37.42847	7.16	0.000	194.2515	341.8707

Decomposition

```
* Generate fitted values and residuals
predict fitted
predict residual, resid
```

```
* Compute averages
sum hhcon fitted residual
```

Variable	Obs	Mean	Std. dev.	Min	Max
hhcon	200	504.1663	329.7687	75.16	1983.02
fitted	200	504.1663	155.9919	268.1639	1022.213
residual	200	-1.28e-07	290.5408	-519.6953	1473.709

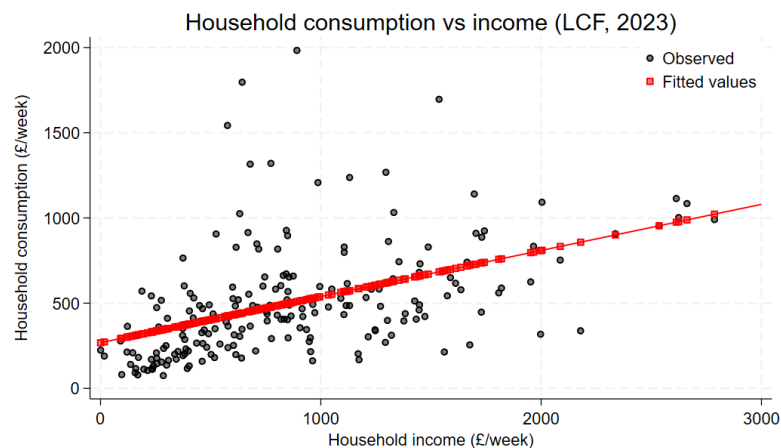
```
* Compute variance-covariance matrix
corr hhcon fitted residual, cov
```

(obs=200)

	hhcon	fitted	residual
hhcon	108747		
fitted	24333.5	24333.5	
residual	84414	-.000076	84414

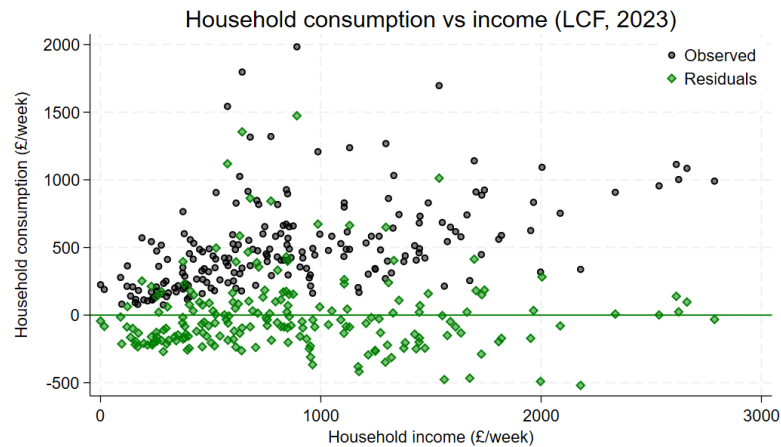
Fitted values

```
* Plot
twoway (scatter hhcon hhinc, mc(black) mfc(%50)) (scatter fitted hhinc, mc(red) mfc(%50) msib)
```



Residual

```
* Plot  
twoway (scatter hhcon hhinc, mc(black) mfc(%50)) (scatter residual hhinc, mc(green) mfc(%50))
```



Binary regressor:

What happens if the regressor is a 'dummy' variable?

$$[hhcons_i, children_i]$$

where $children_i$

- = 1: household has children,
- = 0: household does not have children

Summary stats

```
* Summary table  
tab children, sum(hhcon)
```

Summary of COICOP: Total consumption expenditure - children and adults			
children	Mean	Std. dev.	Freq.
0	481.09786	342.33194	151
1	575.2546	278.91476	49
Total	504.16626	329.76875	200

OLS - dummy variable

```
* Estimate bivariate linear regression
reg hhcon children
```

Source	SS	df	MS	Number of obs	=	200
Model	327978.835	1	327978.835	F(1, 198)	=	3.05
Residual	21312759	198	107640.197	Prob > F	=	0.0824
Total	21640737.9	199	108747.427	R-squared	=	0.0152
				Adj R-squared	=	0.0102
				Root MSE	=	328.09

hhcon	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
children	94.15674	53.94059	1.75	0.082	-12.21506	200.5285
_cons	481.0979	26.69923	18.02	0.000	428.4465	533.7492

You can show,

$$\hat{\beta}_1 = \bar{y}_1 - \bar{y}_0$$

$$\hat{\beta}_0 = \bar{y}_0$$

where,

- $\bar{y}_0$ is the mean of the outcome for observations with $x = 0$

- $\bar{y}_1$ is the mean of the outcome for observations with $x = 1$

Yields a very intuitive interpretation of the OLS estimator

Sample analogue

This is the sample analogue of

$$\beta_1 = E[y_i | x_i = 1] - E[y_i | x_i = 0]$$

$$\beta_0 = E[y_i | x_i = 0]$$

Which was the interpretation of the population parameter in a model with a binary ('dummy') variable.

Multivariate models

What happens if you have two regressors

$$[hhcons_i, hhinc_i, children, hhsiz_e_i]$$

OLS - multivariate

```
* Estimate bivariate linear regression
reg hhcon hhinc children hhsiz_e
```

Source		SS	df	MS	Number of obs	=	200
-----+-----					F(3, 196)	=	27.93
Model		6480977.84	3	2160325.95	Prob > F	=	0.0000
Residual		15159760	196	77345.7144	R-squared	=	0.2995
-----+-----					Adj R-squared	=	0.2888
Total		21640737.9	199	108747.427	Root MSE	=	278.11

hhcon		Coefficient	Std. err.	t	P> t	[95% conf. interval]	
-----+-----							
hhinc		.1937248	.0382801	5.06	0.000	.1182311	.2692185
children		-238.3538	74.84546	-3.18	0.002	-385.9596	-90.74795
hhsiz_e		125.7418	27.57406	4.56	0.000	71.36187	180.1217

_cons		121.2957	47.91901	2.53	0.012	26.79261	215.7987
-------	--	----------	----------	------	-------	----------	----------

Interpretation

Interpretation follows from the fact that,

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \cdots + \hat{\beta}_k x_{ik}$$

$$\hat{\beta}_j = \frac{\partial \hat{y}_i}{\partial x_{ij}}$$

or

$$\Delta \hat{y}_i = \hat{\beta}_j \Delta x_{ij} \quad \text{holding other } x\text{'s fixed}$$

Partialling out

The following result is called the **Frisch-Waugh Theorem** (Wooldridge 2025, 76)

The OLS estimate for $\hat{\beta}_1$ from the multivariate ‘projection’

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \cdots + \hat{\beta}_k x_{ik}$$

is the same as the OLS estimate for $\hat{\gamma}_1$ simple ‘projection’

$$\tilde{\hat{y}}_i = \hat{\gamma}_0 + \hat{\gamma}_1 \tilde{x}_{i1}$$

Projection

Why the phrase ‘projection’? One interpretation of OLS is geometric. The estimator ‘projects’ the outcome (Y) in the (vector) space spanned by the regressors (columns of X).

$$\hat{Y} = X\hat{\beta} = \underbrace{X(X'X)^{-1}X'}_{P_X} Y$$

The predicted value of Y is just a ‘projection’ of Y : $P_X Y$. P_X is an idempotent projection matrix. The residual then is,

$$\hat{U} = Y - \hat{Y} = Y - P_X Y = \underbrace{(I - P_X)}_{M_X} Y$$

M_X is another idempotent projection matrix. And crucially, $M_X P_X = 0$: the two projections are orthogonal. So, OLS projects Y into two vector spaces: one spanned (‘explained by’) X and another orthogonal to that space. The predicted values and residuals are mechanically orthogonal, which means that they are perfectly uncorrelated *in the data*.

where,

- $\tilde{x}_{i1}$ is the residual from the regression of x_1 against all other regressors

$$\tilde{x}_{i1} = x_{i1} - (\hat{\rho}_0 + \hat{\rho}_1 x_{i2} + \dots + \hat{\rho}_{k-1} x_{ik})$$

and,

- $\tilde{y}_i$ is the residual from the regression of y against all regressors *excluding* x_1

$$\tilde{y}_i = y_i - (\hat{\psi}_0 + \hat{\psi}_1 x_{i2} + \dots + \hat{\psi}_{k-1} x_{ik})$$

Demonstration

```
* Estimate bivariate linear regression
qui reg hhinc children hhsz
predict hhinc_tilde, resid
qui reg hhcon children hhsz
predict hhcon_tilde, resid

reg hhcon_tilde hhinc_tilde
```

Source	SS	df	MS	Number of obs	=	200
Model	1980895.01	1	1980895.01	F(1, 198)	=	25.87
Residual	15159759.9	198	76564.4441	Prob > F	=	0.0000
				R-squared	=	0.1156
				Adj R-squared	=	0.1111

Total | 17140654.9 199 86133.9444 Root MSE = 276.7

hhcon_tilde	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
hhinc_tilde	.1937248	.0380862	5.09	0.000	.1186181	.2688315
_cons	-2.04e-08	19.56584	-0.00	1.000	-38.58418	38.58418

Goodness-of-Fit

How well did we fit the data?

We have decomposed each observation into two parts:

$$y_i = \underbrace{\hat{y}_i}_{\text{fitted values}} + \underbrace{\hat{u}_i}_{\text{residual}}$$

We can apply this decomposition to the *total sum of squares* (SST)

$$\text{SST} \equiv \sum_{i=1}^n (y_i - \bar{y})^2$$

- The SST is a measure of the total variation in y .

We can show that (see Wooldridge 2025, 33–34),

$$\text{SST} = \text{SSE} + \text{SSR}$$

where,

- *explained sum of squares* (SSE):

$$\text{SSE} \equiv \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

- *residual sum of squares* (SSR):

$$\text{SSR} \equiv \sum_{i=1}^n \hat{u}_i^2$$

Measure of fit

One key measure of fit then is:

$$\mathbf{R}^2 = \frac{\text{SSE}}{\text{SST}} = 1 - \frac{\text{SSR}}{\text{SST}}$$

This measure captures the share of the variance in the outcome variable explained by the regressors.

- If $\mathbf{R}^2 = 1$ then regressors perfectly predict the outcome.

Chasing $\mathbf{R}^2$

Many students make the mistake of thinking that higher $\mathbf{R}^2$ is always better. Here are reasons this need not be the case:

1. In social sciences, the DGP is often very complex and we don't observe many explanatory variables.
 - given the same set of $\mathbf{x}$'s, different outcomes may have very different $\mathbf{R}^2$
2. If we care about the relationship between y and a specific x_j , we do not necessarily care about explaining all of y .
3. We can always add variables and (mechanically) increase $\mathbf{R}^2$; however, this comes at a potential cost:
 - a. we *MAY* inadvertently increase the variance of our estimates (e.g., *irrelevant* variables)
 - b. we *MAY* end up adding “bad controls”

Adjusted R^2

An alternative measure fit is the *adjusted* R^2 :

$$\overline{R}^2 = 1 - \frac{SSR/(n-k-1)}{SST/(n-1)} = 1 - (1 - R^2) \times \frac{(n-1)}{(n-k-1)}$$

where,

- n is sample size
- k is the number of regressors $\Rightarrow k+1$ is the number of parameters estimated (including intercept).

As you increase k , you decrease $\overline{R}^2$

- Creates a disincentive to add irrelevant regressors that do not also reduce SSR .

Example: LCF (2023)

.29948045

.28875821

Adding “irrelevant” variables

```
reg hhcon hhinc hhsize children month tvlicense
```

Source		SS	df	MS	Number of obs	=	200
-----+					F(5, 194)	=	16.91
Model		6568404.46	5	1313680.89	Prob > F	=	0.0000
Residual		15072333.4	194	77692.4403	R-squared	=	0.3035
-----+					Adj R-squared	=	0.2856
Total		21640737.9	199	108747.427	Root MSE	=	278.73

hhcon		Coefficient	Std. err.	t	P> t	[95% conf. interval]	
-----+							
hhinc		.1911845	.0385338	4.96	0.000	.1151854	.2671835
hhsize		124.161	27.71213	4.48	0.000	69.50531	178.8168
children		-233.0088	75.21666	-3.10	0.002	-381.3562	-84.66142

month		-1.276233	6.234806	-0.20	0.838	-13.57294	11.02047
tvlicense		59.82271	56.57194	1.06	0.292	-51.7523	171.3977
_cons		82.86154	77.50891	1.07	0.286	-70.00675	235.7298

Does OLS work?

Are these the “right” numbers?

Source		SS	df	MS	Number of obs	=	200
					F(1, 198)	=	57.08
Model		4842359.39	1	4842359.39	Prob > F	=	0.0000
Residual		16798378.5	198	84840.2954	R-squared	=	0.2238
					Adj R-squared	=	0.2198
Total		21640737.9	199	108747.427	Root MSE	=	291.27

hhcon		Coefficient	Std. err.	t	P> t	[95% conf. interval]	
hhinc		.2705895	.0358165	7.55	0.000	.1999587	.3412203
_cons		268.0611	37.42847	7.16	0.000	194.2515	341.8707

Let’s evaluate the question:

1. it pre-supposes that there is a *right* answer (i.e. a correct estimate)
 - i.e., a *true* population parameter (β)
2. it implies a notion of “correctness”
 - do we mean $\beta = 0.27$?
 - do we mean $\beta \approx 0.27$?
 - do we mean β is ‘close to’ 0.27?
3. it assumes that the OLS estimator can be *right*.

First off, let's clarify something:

$$\hat{\beta}$$

- the estimator

...is a random variable

$$0.27$$

- the estimate

...is just a number

Accuracy

Is $\hat{\beta} = \beta$?

- No!
- In fact, the statement does not make sense since $\hat{\beta}$ is a random variable while β is a non-random constant.
- If we are accurate, we can ask: $Pr(\hat{\beta} = \beta) = ?$
- The answer is: $Pr(\hat{\beta} = \beta|X) = 0$ for any u with a continuous distribution (conditional on X : the sample of values of independent regressors).

Random variable

In what sense is $\hat{\beta}$ a random variable?

- The realized value of $\hat{\beta}$ depends on the realized sample¹
- In another 'state of the world', the sample may have realized differently

¹Recall, if u is independent of $\mathbf{x}$, then we can treat $\mathbf{x}$ as fixed. Therefore, the randomness (across samples) is really from different realizations of u . However, any discussion of u and $\mathbf{x}$ pre-supposes a model which we have not yet defined.

Example: LCF (2023) - Version 2

I randomly sampled 200 observations from the LCF (2023) data, but could have equally have sampled a different 200 observations

Source	SS	df	MS	Number of obs	=	200
				F(1, 198)	=	90.01
Model	5283929.28	1	5283929.28	Prob > F	=	0.0000
Residual	11623093.9	198	58702.4945	R-squared	=	0.3125
				Adj R-squared	=	0.3091
Total	16907023.2	199	84959.9155	Root MSE	=	242.29

hhcon	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
hhinc	.2712055	.0285857	9.49	0.000	.2148341	.327577
_cons	232.6529	30.55968	7.61	0.000	172.3886	292.9171

Example: LCF (2023) - Version 3

... and again,

Source	SS	df	MS	Number of obs	=	200
				F(1, 198)	=	102.09
Model	7462064.65	1	7462064.65	Prob > F	=	0.0000
Residual	14472533.5	198	73093.6037	R-squared	=	0.3402
				Adj R-squared	=	0.3369
Total	21934598.2	199	110224.111	Root MSE	=	270.36

hhcon	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
hhinc	.3461268	.0342567	10.10	0.000	.278572	.4136816
_cons	189.5292	36.88368	5.14	0.000	116.794	262.2645

Example: LCF (2023) - Version 4

... and again.

Source	SS	df	MS	Number of obs	=	200
				F(1, 198)	=	27.79
Model	3538499.21	1	3538499.21	Prob > F	=	0.0000
Residual	25213603.4	198	127341.431	R-squared	=	0.1231
				Adj R-squared	=	0.1186
Total	28752102.6	199	144482.928	Root MSE	=	356.85

hhcon	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
hhinc	.241712	.0458536	5.27	0.000	.1512879	.3321361
_cons	267.7737	48.63081	5.51	0.000	171.8729	363.6745

Can we **expect** – *a priori* – for $\hat{\beta}$ to be “on target”?

- This is a question of *biasedness*: is $E[\hat{\beta}] = \beta$?
- The answer: it depends!
- Specifically, it depends on the underlying **model**!

Note

It is an all too common mistake to conflate the estimator - in particular, the OLS estimator - with the linear regression model. People will say, “We estimate an OLS regression model”. This doesn’t really make sense, since you OLS is the estimator, not the name of the model. You can say, “We estimate a linear regression model using OLS”.

This may seem pedantic, but it is an important distinction since there are a range of estimators that can be used to estimate a linear regression model; all with different properties. In addition, the OLS estimator can be applied to any data, regardless of the true model. So it is a good idea to make sure your language is accurate.

References

Bibliography

Wooldridge, Jeffrey M. 2025. “Introductory Econometrics: A Modern Approach.”